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Decodelabs Internship

Data Cleaning, Preprocessing & Visualization Analysis

Retail Store Sales

Executive Summary

This report documents the complete data analysis pipeline for retail store sales data, including data cleaning and preprocessing using Python's Pandas library, followed by Power BI visualizations. The project demonstrates essential data engineering practices to prepare raw data for business intelligence.

Dataset Overview

Data Structure

The retail store sales dataset contains transaction-level information with the following columns:

Column Name	Data Type	Description
Transaction Date	DateTime	Date of transaction
Item	String	Product name
Price Per Unit	Float	Unit price of item
Quantity	Integer	Quantity purchased
Total Spent	Float	Total transaction value
Discount Applied	Boolean	Discount flag

Data Cleaning & Preprocessing

- Loaded the dataset using Pandas read_csv() function
- Examined dataset structure using head(), info(), and shape()
- Identified data types and potential issues
- Checked for missing values and duplicate records

Missing Value Treatment

Missing values were handled using appropriate strategies based on data type and business logic:

Column	Missing Value Treatment	Reason
Item	Filled with 'Unknown'	Categorical column requiring label value
Price Per Unit	Filled with median value	Median is robust to outliers in price data
Quantity	Filled with median value	Preserves central tendency
Total Spent	Filled with median value	Median minimizes impact of transaction extremes
Discount Applied	Filled with mode value	Mode preserves most common discount pattern

Duplicate Records

Checked for duplicate rows using `.duplicated()` method. Duplicates were not identified in the dataset and decided to move on to the next step.

Data Type Conversion

Critical data type conversions were performed:

- Transaction Date: Converted to datetime format using `pd.to_datetime()` with `dayfirst=True` parameter (appropriate for DD-MM-YYYY format)
- Discount Applied: Converted to boolean type for efficient binary representation and comparison operations

Detailed Code Explanation

Data Loading

The dataset is loaded into a Pandas DataFrame using the `read_csv()` function. This creates an in-memory tabular structure for manipulation and analysis.

Data Exploration

Three key exploration methods:

- **`df.head()`:** Returns first 5 rows to visually inspect data
- **`df.info()`:** Displays data types, non-null counts, and memory usage for all columns
- **`df.shape`:** Returns tuple of (rows, columns) to understand dataset dimensions

Missing Values & Duplicates

`df.isnull().sum()`: Creates a count of missing values per column

`df.duplicated().sum()`: Identifies completely duplicate rows (same values across all columns)

Data Imputation Strategy

Missing values are imputed using column-specific strategies:

Median Imputation (Numeric columns): Used for Price Per Unit, Quantity, and Total Spent because median is resistant to outliers and maintains statistical properties better than mean for skewed distributions.

Mode Imputation (Categorical/Boolean): Used for Discount Applied to fill with the most frequently occurring discount status, preserving the distribution pattern.

Type Casting

Transaction Date Conversion:

`pd.to_datetime(df['Transaction Date'], dayfirst=True)` converts string dates to datetime objects.

The `dayfirst=True` parameter ensures proper parsing of DD-MM-YYYY format dates, critical for accurate time-series analysis.

Boolean Conversion:

`astype(bool)` converts Discount Applied column to boolean type, reducing memory usage and improving filter/comparison operations in Power BI.

Final Output

The cleaned dataset is exported to `retail_store_sales_cleaned.csv` using `df.to_csv()` with `index=False`, ensuring no unnecessary index column is saved. This cleaned file becomes the source for Power BI visualization.

Power BI Visualization Analysis

Visualization Overview

Power BI visualizations have been created based on the cleaned retail sales data. The following section describes the key visualizations and their insights. Please refer to the Power BI report file for the interactive dashboard.

Key Visualizations

Create charts and graphs are selected as below:

Sales Over Time - **Line/Area chart** showing transaction trends and seasonality

Top Products - **Bar chart** highlighting best-selling items by quantity or revenue

Discount Impact Analysis - Comparison of discounted vs non-discounted transactions

Price Distribution - Histogram or distribution chart of product prices

Key Metrics - KPI cards displaying total revenue, average transaction value, and discount percentage

Key Insights

Sales Trends

TIME SERIES ANALYSIS: The transaction date visualization reveals patterns in customer purchasing behavior over time. This enables identification of peak sales periods, seasonal trends, and anomalies. Understanding these patterns helps optimize inventory management and marketing campaigns.

Product Performance

PRODUCT RANKING: Analysis of the Item column reveals which products drive the most revenue and volume. This information guides purchasing decisions and helps identify slow-moving inventory that may need promotional intervention.

Discount Effectiveness

PROMOTIONAL IMPACT: The Discount Applied boolean field enables comparison between discounted and regular-price transactions. Key questions: Do discounts increase transaction

volume? What is the impact on total revenue? Is there a correlation between discounts and specific product categories?

Revenue Metrics

TRANSACTION VALUE ANALYSIS: Total Spent combined with Price Per Unit and Quantity provides complete revenue visibility. Average transaction value, revenue per product, and customer lifetime value can be calculated, revealing high-margin opportunities and customer segments worth targeting.

Data Quality Improvements

CLEANED DATA IMPACT: The data cleaning process removed inconsistencies and standardized formats, ensuring Power BI calculations are accurate. This is reflected in reliable metric calculations and trustworthy trend analysis.

Summary & Recommendations

Key Achievements

- Successfully cleaned and standardized raw retail sales data using Pandas
- Implemented appropriate missing value imputation strategies based on data characteristics
- Converted data types for efficient processing and analysis
- Created Power BI visualizations enabling business stakeholder insights
- Generated production-ready cleaned dataset for ongoing analysis

Recommendations

- Objectively analyze the data after setting up automatic data validation workflows to detect problems with data quality.
- For large data sets with complex patterns, think about using some data imputation techniques: KNN, iterative imputation.
- Configure periodic refreshes of Power BI dashboards for up-to-the-minute insights
- Set alerts for KPI thresholds and take proactive steps to address changes in the business
- Describe a data dictionary and revise as additional variables are added

Conclusion

The entire process of data analytics, from cleaning data, identifying value insights, to viewing the value insights in a business intelligence way, was demonstrated in this project. The methodical approach to dealing with missing values, data type standardization, and making use of visualizations to take action is the base for decision making based on data in retail operations. These cleaned data and Power BI dashboards help stakeholders pinpoint trends, optimize product range and assess the performance of their promotional efforts, without spending hours on data analysis.